

Kalshi Order-Book Predictor Project

Predicting Short-Term Price Movement on KXBTC15M via Order Book Imbalance

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1. Background

Kalshi is a CFTC-regulated U.S. prediction market exchange that lists more than 700 event-based binary contracts. Each contract trades between \$0.01 and \$0.99 and settles at \$1.00 if the underlying event occurs and \$0.00 otherwise. Crucially, every trade on Kalshi is between two retail or institutional participants rather than against the exchange itself, so the limit order book is the entire market and there is no designated market maker subsidizing liquidity. Because Kalshi attracts a heavy retail user base that predominantly takes liquidity through market orders, the remaining book is largely populated by a smaller pool of profit-driven liquidity providers. This produces a structurally interesting microstructure: a handful of informed makers posting size at multiple levels can leave detectable footprints before the best bid/ask shifts.

A Kalshi contract is naturally symmetric: the YES price plus the NO price is always \$1.00. Buying YES at \$0.65 is mathematically identical to selling NO at \$0.35, and the WebSocket feed reflects this directly — only `yes_bids` and `no_bids` are streamed, and the asks have to be reconstructed in memory using the identity $\text{yes_ask} = 100 - \text{no_bid}$ (and vice versa). One welcome consequence is that there is no shorting problem: directional exposure on either side of a contract is acquired by simply buying the appropriate ticker.

We chose to focus on KXBTC15M, the rolling “BTC up or down in 15 minutes” market. A new contract opens every 15 minutes (e.g. KXBTC15M-26MAR042130-30 covers the window ending at 21:30 UTC), and settlement is determined by the average of 60 CF Benchmarks BTC RTI prices in the final minute. The market clears around \$170k of notional volume per window, with on the order of 100 order book updates per second during active periods — roughly two orders of magnitude busier than most other Kalshi contracts. The combination of high tick rate, a single, well-defined external reference price (BTC spot), and zero maker fees on this specific market made it the obvious candidate for an HFT-style microstructure study. We deliberately avoided sports markets where adverse selection is dominated by fundamental sports knowledge that we do not have.

Fees are non-trivial and shape every strategy decision. Taker fees are computed as $\text{round-up}(0.07 \times C \times P \times (1-P))$ per contract, where C is contract count and P is the fill price. The fee is therefore highest at \$0.50 (~\$0.0175 per contract, rounded up to \$0.02) and shrinks toward the tails. With a \$0.01 tick and contract values capped at \$1.00, a single tick is roughly 1–2% of the underlying. Any signal that fires in the middle of the price range therefore has to predict at least a 2–3 tick move just to break even on round-trip taker fees — a high bar for a tick-level signal.

2. Data Collection and Initial Framework

To collect tick-level data we wrote a multi-threaded C++ collector that connects to Kalshi over WebSocket and to Coinbase’s ticker stream in parallel. A coordinator thread reads the local clock, computes the next KXBTC15M ticker symbol from UTC time, and spins up a dedicated worker thread several minutes before each market opens so that the very first orderbook_snapshot of the session is captured. Each worker subscribes to that single market’s channel, reconstructs an in-memory

price-by-quantity book from the `orderbook_delta` stream (treating every level as a 99-slot dense vector indexed by cents), and writes one row per book change to disk. Because Kalshi only sends YES/NO bid sides, the asks are computed live in C++ rather than being read from the wire. We stored data in flat CSVs partitioned by market ticker; one 15-minute window weighed in at 15–35 MB of Kalshi data plus 5–7 MB of Coinbase ticker, or roughly 2.5–3 GB per trading day.

For BTC reference data we deliberately chose Coinbase’s ticker channel rather than full L2. The price that determines KXBTC15M settlement is the CF Benchmarks RTI, which is itself a price aggregate, so resting book depth on Coinbase carries no marginal information beyond the latest traded price. Using ticker+market trades cut Coinbase bandwidth by roughly 10x relative to L2 with no loss of signal. We over-recorded the BTC stream by a few minutes on each side of every Kalshi window and trimmed the buffer at analysis time to absorb any clock skew between feeds.

The collector ran 24/7 from a campus lab machine. After about a week of continuous operation the host crashed under disk pressure and we lost a small amount of unflushed data; we restarted with rotated logs and rebuilt our dataset from there. By the end of the data-gathering phase we had several hundred fully captured KXBTC15M windows along with synchronized BTC ticker data — enough to support both signal research and held-out backtesting.



Figure 1. Reconstructed market replay for a single KXBTC15M window. Top: YES mid-price with best bid/ask. Middle: synchronized BTC spot (Coinbase ticker). Bottom: OBI(depth=5, decay=0.2) derived from the reconstructed book.

All downstream analysis was performed in Python (Pandas, scikit-learn, XGBoost) reading the CSVs directly. The raw representation — a per-tick 99-dimensional depth vector plus best bid/ask plus matched BTC ticker — was permissive enough that we never had to recollect data when iterating on signal definitions.

3. OBI Delta

Our first hypothesis was direct: sudden, large changes in order book imbalance precede short-horizon price moves. The intuition is that informed flow shows up as a burst of orders appearing on one side of the book a fraction of a second before the mid moves. We computed OBI as the depth-weighted ratio across the top five levels:

$$OBI = (yes_weighted - no_weighted) / (yes_weighted + no_weighted),$$

with exponential level weights (decay=0.2) so that the top of the book dominates and stale, deep liquidity contributes little. We then defined the delta signal d_obi as $OBI(t) - OBI(t - 0.25\text{ s})$ and fired a buy when $d_obi \geq +0.4$ (and a sell when $d_obi \leq -0.4$). Forward returns were measured at a one-second horizon with a minimum tick move of \$0.01, and a 0.5-second cooldown enforced between firings to avoid double-counting bursts.

OBI Delta Signal Test	
OBI column	: obi (depth=5, decay=0.2)
Signal window:	0.25s Horizon: 1s
Threshold	: d_obi >= 0.4
Min tick move	: 0.01
Cooldown	: 0.5s
Total signals : 498	
BUY (d_obi > 0)	: 255
SELL (d_obi < 0)	: 243
Hit rate	: 39.0% (194/498)
Adverse rate	: 20.3% (101/498)
No-move rate	: 40.8% (203/498)
Avg fwd move	: +0.00245 (all signals)
Avg hit move	: +0.01247 (hits only)
Avg adv move	: -0.01188 (adverse only)
BUY hit rate	: 41.2% (n=255)
SELL hit rate	: 36.6% (n=243)

Figure 2. OBI delta signal test results across one tagged session. The hit rate is barely above no-move, and the average forward move on hits (\$0.0125) is less than two ticks — not enough to clear taker fees on round-trip taking.

The aggregate statistics across 498 firings were 39.0% hit rate, 20.3% adverse, and 40.8% no-move, with an average forward move of +\$0.00245 across all firings and +\$0.01247 conditional on a hit. The signal does have a small directional edge — hits outnumber adverses two-to-one — but the magnitude is the problem. A taker round-trip in the middle of the price range costs about two ticks (\$0.02) in fees alone, so even a clean hit at the average magnitude returns roughly zero net. The signal scales the wrong way: making the threshold more selective reduces firing rate without meaningfully improving per-trade payoff, because the underlying move distribution is too tight relative to the tick size.



Figure 3. Per-tick OBI signal annotations on a representative session. Buy and sell signals are color-coded; no-move and adverse signals are marked separately. Visual inspection confirms that hits and adverses cluster together rather than being temporally separable.

We treat the OBI delta result as evidence that the directional information in the book is real but too small to extract via aggressive market taking on this market. That conclusion is what motivated the next two attempts — first a more sophisticated calibrated signal (microprice), and later a switch from taking to making.

4. Microprice Calibration

The microprice is a calibrated correction to the mid-price that accounts for short-term order book imbalance. Following the standard formulation, it models the joint dynamics of (M, S, I) — mid-price, spread, and volume imbalance — as a three-dimensional Markov process and learns a lookup table of bias adjustments $G^*[\text{imbalance bucket}, \text{spread bucket}]$ from historical transition counts. The signal is defined as

$$\text{microprice} = \text{mid} + G^*[\text{imbalance bucket}, \text{spread bucket}],$$

and a trade is suggested whenever $|\text{microprice} - \text{mid}|$ crosses some threshold. We bucketed imbalance into 10 bins and spread into 3 bins (1, 2, and 3+ ticks) and calibrated G^* on roughly 2 million transitions across our dataset.

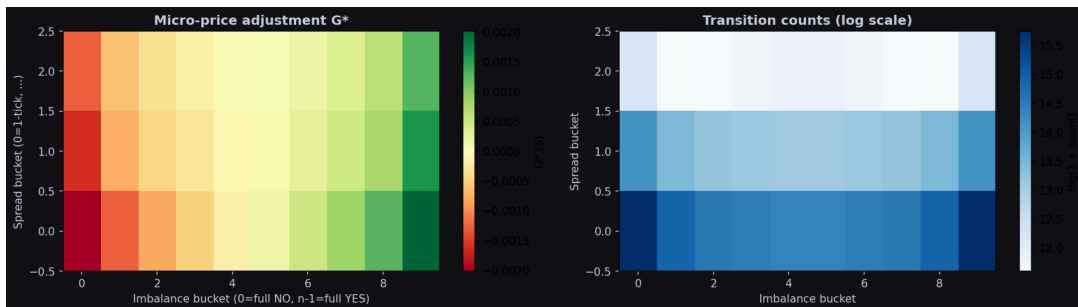


Figure 4. Calibrated microprice adjustment G^ (left) and log-transition counts (right) across imbalance and spread buckets on KXBTC15M. The shape is qualitatively correct — G^* leans positive when YES depth dominates and the spread is tight — but the magnitude is everywhere bounded by $\sim \pm \$0.0022$.*

The qualitative shape of the calibrated table is reasonable: G^* is positive when imbalance leans YES and the spread is tight, and it shrinks toward zero as the spread widens or imbalance weakens. The fatal problem is one of magnitude. The maximum absolute value of G^* anywhere in the calibrated table is approximately \$0.0022 — less than one quarter of a tick on a \$0.01 grid. There is no actionable price level between mid and microprice that an order can rest at, so the signal cannot translate into a real fill at a real price. Lowering the firing threshold to make signals appear at all just degrades into noise trading at sub-tick magnitudes.

Stepping back, this is not an implementation bug; it is a structural mismatch between the model and the market. Microprice is designed for instruments where ticks are fine relative to spread (e.g. equities at \$0.0001 ticks), so a sub-cent G^* maps onto multiple actionable price levels. On a binary contract priced in cents between \$0 and \$1, the entire actionable price range is only 99 ticks wide and the natural scale of the imbalance correction is finer than the grid. We stopped pursuing microprice as a live signal at this point and removed it from the active signal registry, while keeping the calibration code for future markets where the tick size is friendlier.

5. Ensemble Model

With OBI deltas too small to take through fees and microprice off the grid, we tried fitting a more flexible model: a gradient-boosted classifier that predicts whether the YES mid will be higher 10 seconds in the future. We engineered 16 features spanning depth, imbalance, and BTC dynamics: best YES/NO bid prices and quantities, multi-level OBI (`obi_1`, `obi_3`, `obi_5` with exponential decay), book elapsed time within the 15-minute window, total top-of-book size, and three BTC-side features — `vol_imbalance` over a 15-minute rolling window, and BTC price momentum over 5- and 10-second windows. Training used roughly 150 sessions held back from the held-out test set; the target was a binary 10-second forward direction label.

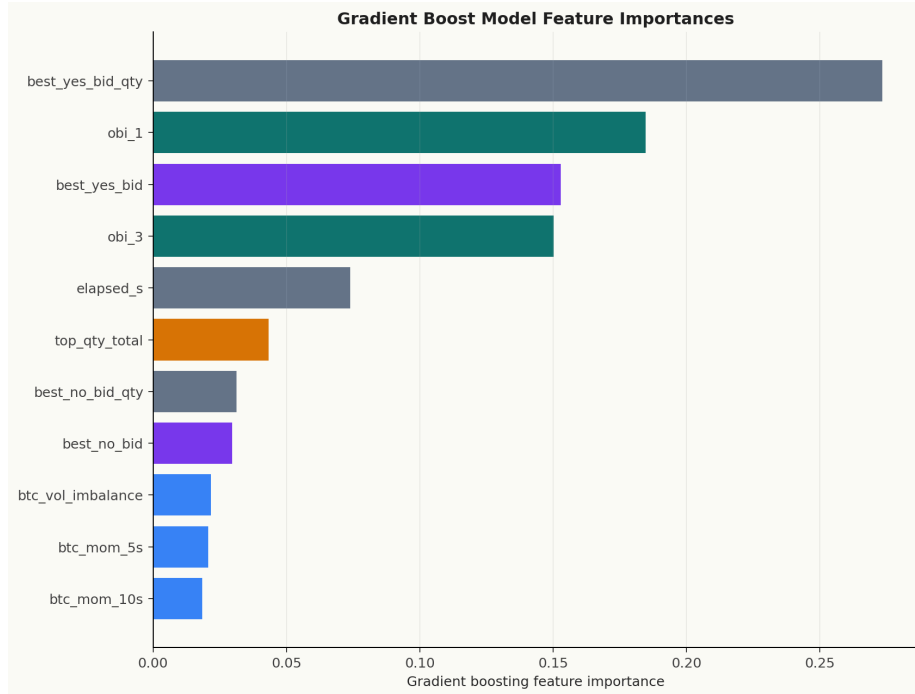


Figure 5. Gradient-boosted feature importances. Top-of-book quantity and obi_1/obi_3 dominate the model; explicit BTC momentum features carry residual importance but are not the largest contributors.

In-sample, the model reached an information coefficient of roughly 0.28 against the 10-second forward return and AUC ~0.67 — enough to demonstrate that the signal is real, but not large by HFT standards. The natural temptation was to convert this directly into a trading rule: post passive YES quotes when $P(\text{up}) > 0.55$ and passive NO quotes symmetrically. On the bench mark-to-mid simulation the strategy showed strong P&L — but that simulation assumes fills at the mid-price on both legs, which is impossible in practice. The honest version of the simulator, which crosses the spread to enter and exit, gives back essentially all of the apparent edge.

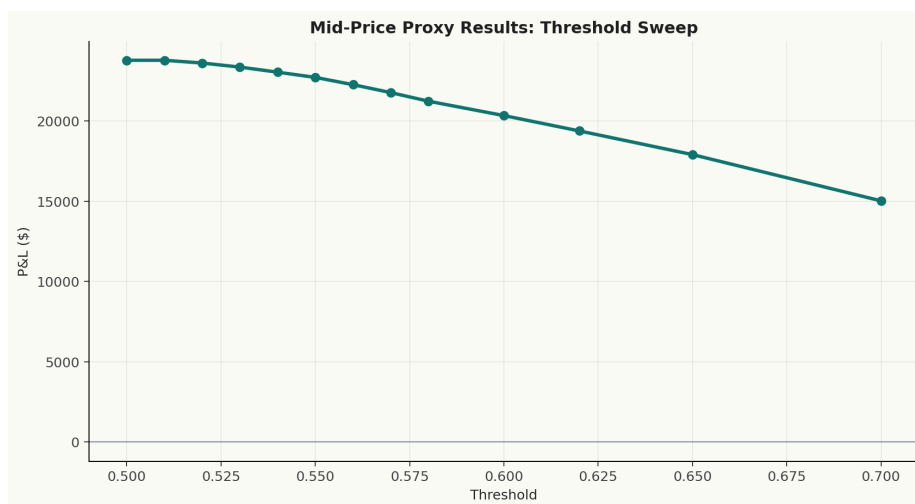


Figure 6. P&L of the GBM “mid-price proxy” simulator across thresholds. The P&L is large and monotone in the threshold — a hallmark of an unrealistic fill model rather than a real edge.

The mid-price simulator is essentially measuring sign accuracy, not tradable edge. Once we forced the simulator to either pay the spread on entry/exit (taking) or wait for someone to come hit our quote (making), the picture changed dramatically. As a market-maker entry filter the GBM at threshold 0.55 produced a 17% win rate, near-zero net P&L, and very few firings during quiet overnight sessions — precisely the regime where the rule-based OBI strategy still finds edge. Two structural problems explain the gap. First, the model predicts a 10-second horizon while a passive quote that gets filled is filled in 100–500 milliseconds; by that point the imbalance the model conditioned on has decayed and the order has been picked off by a taker who, by virtue of being willing to cross the spread, holds the better short-horizon view. This is adverse selection in its textbook form. Second, the model was trained on March data and evaluated on April data, and the BTC volatility regime shifted enough between the two months that the conditional distributions drifted; the unconditional AUC of 0.67 hides the fact that the conditional accuracy given a fill is much lower.

The takeaway from the ensemble experiment is that the directional signals exist but converting a directional prediction into a market-making strategy throws away the primary source of market-maker edge — spread capture and adaptive cancellation — in exchange for a probabilistic entry filter whose probability quickly decorrelates from the post-fill state. This pushed us toward a much simpler rule-based formulation.

6. A Simpler Rule-Based Market Maker

After the ensemble experiment we redesigned the strategy from the perspective of a market maker, not a forecaster. The premise is that on KXBTC15M we earn the spread on fills, pay zero maker fees, and only need a directional filter strong enough to keep us from being adversely selected. The filter doesn't need to predict price; it just needs to tell us when not to quote.

The resulting strategy uses two independent agreements that must both hold to post a quote:

- Post a YES bid at $\text{best_bid} + 1\text{¢}$ if $\text{OBI} > +0.05$ AND $\text{btc_mom_10s} \geq 0$.
- Post a NO bid (equivalently, a YES ask at $\text{best_ask} - 1\text{¢}$) if $\text{OBI} < -0.05$ AND $\text{btc_mom_10s} \leq 0$.
- Cancel and force-exit if $|\text{btc_mom_10s}| > 0.0001$ against an open position (a 0.01% BTC move in the wrong direction over 10 s).

OBI here is the level-1 reading $(\text{best_yes_bid_qty} - \text{best_no_bid_qty}) / (\text{best_yes_bid_qty} + \text{best_no_bid_qty})$, a memoryless contemporaneous measurement; btc_mom_10s is the percentage change in Coinbase BTC over the last 10 seconds. The entry rule looks only at the sign of BTC momentum (no minimum magnitude required), but the cancel rule uses a magnitude threshold so that a sustained BTC move against an open position pulls us out before it reaches the Kalshi mid.

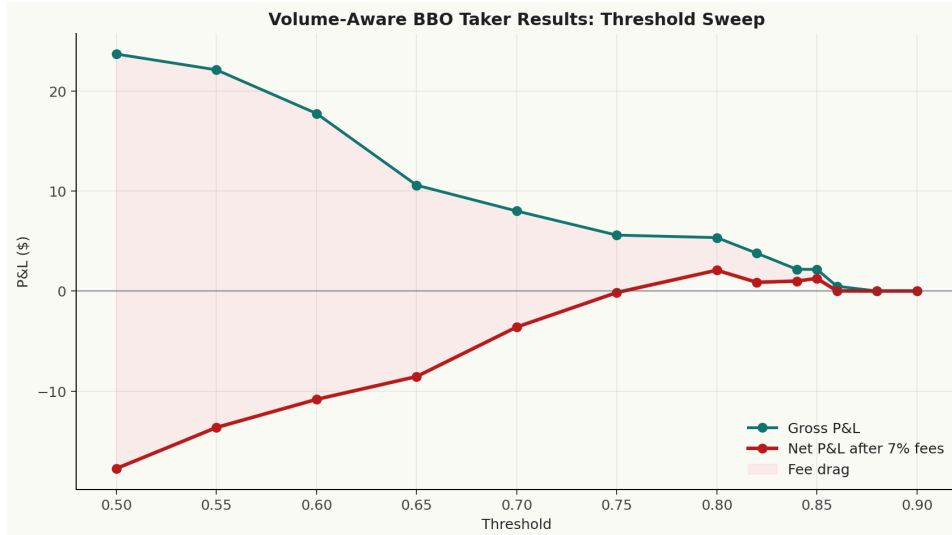


Figure 7. Volume-aware best-bid/offer taker variant for comparison: gross P&L is positive across most thresholds but is consumed entirely by the 7% taker fee at every operating point. The equivalent maker strategy avoids this fee drag entirely.

Run across a 114-session held-out test set, the rule-based market maker achieves a 91% win rate and +\$40 net P&L at the best-tuned operating point (OBI threshold 0.05, qty=25), versus the GBM-based passive strategy at +\$3 across the same windows. The GBM and naive strategies are genuinely different families: the GBM occasionally captures larger discrete wins when its probability estimate spikes, while the rule-based strategy delivers a steadier accumulation of small spread-capture P&L.

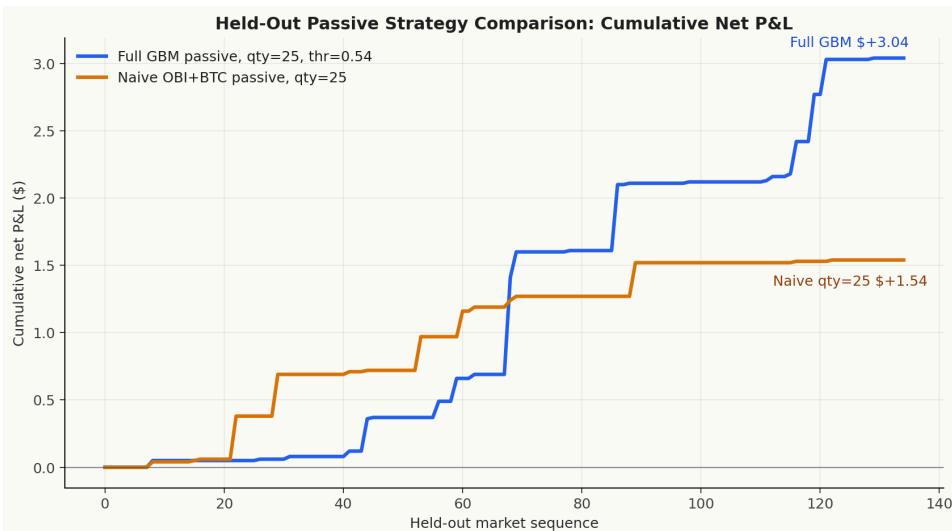


Figure 8. Cumulative net P&L on the held-out market sequence: the full GBM passive ends at +\$3.04, the naive OBI+BTC passive at +\$1.54 at smaller quote sizes. At qty=25 the rule-based strategy reaches roughly +\$40, with the bulk of the lift coming from longer holds and broader entry conditions.

A separate ablation — a depth-skew variant where quote prices were tilted based on the same OBI signal — added nothing on top of the basic threshold rule. The interpretation is that the edge here is essentially a liquidity premium that the resting maker charges in exchange for the option of posting only when the

book already leans in their direction. The strategy is not making a directional bet so much as renting capital to retail takers under a filter that excludes the worst-conditioned moments.

7. Live Testing

To validate the rule-based strategy out of sample we extended the C++ collector into a live trading client. Market data continues to arrive over WebSocket (orderbook_snapshot, orderbook_delta, fill, trade), and order placement and cancellation are issued via synchronous HTTPS REST calls to Kalshi's portfolio endpoint. Each order incurs roughly 100–200 ms of network and TLS overhead per call; at the order rates we run, keep-alive complexity wasn't worth it. Orders are tagged Post-Only via the API flag so that we can never accidentally take liquidity and pay fees while a quote is being placed across a moving market.

```
[MM:fill] ENTRY dir=-1 price=0.53 qty=2
[REST] DELETE /trade-api/v2/portfolio/orders/d2c30fff-8820-49e7-9f33-ba13df4410f1 → HTTP 404: {"error":{"code":"not_found","message":"not found","service":"exchange"}}
[MM:live] cancel 404: d2c30fff-8820-49e7-9f33-ba13df4410f1 was filled — waiting for fill channel
[MM:fill] EXIT price=0.52 net=$0.02 hold=1.9s
[MM] bid=0.50 ask=0.51 mid=0.51 OBI=0.956 btc_mom=-0.0268% BTC=$77651 | FLAT | trades=1 win=100.0% net=$0.02 | skip=154 cancel=3
[MM:live] placed buy yes 50¢ x2 order_id=f3689f3a-541b-4d28-9d4f-edccc316b81
[MM:live] cancelled f3689f3a-541b-4d28-9d4f-edccc316b81
[MM:live] placed buy yes 50¢ x2 order_id=c2aec940-20ba-4365-bc93-598b08a60362
[MM:live] cancelled c2aec940-20ba-4365-bc93-598b08a60362
[MM:live] placed buy no 47¢ x2 order_id=a7db9787-30e0-424e-ab40-1e54e34326bd
[MM:live] cancelled a7db9787-30e0-424e-ab40-1e54e34326bd
[MM] bid=0.51 ask=0.52 mid=0.52 OBI=0.985 btc_mom=-0.0038% BTC=$77651 | FLAT | trades=1 win=100.0% net=$0.02 | skip=193 cancel=6
[MM:live] placed buy no 48¢ x2 order_id=15ce8997-c892-4ed0-a902-dd0064748c50
[MM:live] cancelled 15ce8997-c892-4ed0-a902-dd0064748c50
[MM] bid=0.50 ask=0.51 mid=0.51 OBI=0.973 btc_mom=-0.0067% BTC=$77645 | FLAT | trades=1 win=100.0% net=$0.02 | skip=193 cancel=7
[MM:live] placed buy no 49¢ x2 order_id=53e46bda-8179-46b1-9709-10dd4f86b191
[MM] bid=0.49 ask=0.50 mid=0.49 OBI=0.969 btc_mom=-0.0051% BTC=$77647 | POSTING ASK 0.51 | trades=1 win=100.0% net=$0.02 | skip=193 cancel=7
[REST] DELETE /trade-api/v2/portfolio/orders/53e46bda-8179-46b1-9709-10dd4f86b191 → HTTP 404: {"error":{"code":"not_found","message":"not found","service":"exchange"}}
[MM:live] cancel 404: 53e46bda-8179-46b1-9709-10dd4f86b191 was filled — waiting for fill channel
[MM:live] placed sell no 51¢ x2 order_id=e8c8bc14-5a5d-45a5-a811-ee9d3f12baa1
[MM:fill] ENTRY dir=-1 price=0.51 qty=2
[MM:live] cancelled e8c8bc14-5a5d-45a5-a811-ee9d3f12baa1
[MM:live] placed sell no 50¢ x2 order_id=d57c5149-5ec3-4378-b877-ca85482d0f41
[MM] TAKER EXIT price=0.50 fee=$0.0400 net=$-0.02
```

Figure 9. Excerpt from the live MM session log. Each line records a quote placement, cancellation, fill, or REST response from Kalshi, with the corresponding mid, OBI reading, and BTC momentum that triggered (or canceled) the action.

We turned the live system on for approximately six hours of intraday operation with a \$2 starting bankroll and walked away with roughly \$1.50 of net profit. As a paper-thin sample the result is encouraging — the strategy made positive money in production — but the operational lessons matter more than the headline number. Most strikingly, the resting bid/ask spread on KXBTC15M had compressed dramatically since our March data collection: at multiple points during the live session there was no bid-ask spread to capture at all, and the strategy was reduced to either crossing or not quoting. This is consistent with additional makers entering the market over the intervening two months and is the single biggest threat to the long-run viability of the approach.

Yes · Target Price: \$77,790.03	Cost \$ \$3.18	Apr 23 · Paid out \$3.24	▼
 BTC 15 min · \$77,728.27 target			
No · Target Price: \$77,728.27	Cost \$ \$0.84	Apr 23 \$0	▼
 BTC 15 min · \$77,601.02 target			
No · Target Price: \$77,601.02	Cost \$ \$0.94	Apr 23 \$0	▼
 BTC 15 min · \$77,492.26 target			
Yes · Target Price: \$77,492.26	Cost \$ \$5.08	Apr 23 · Paid out \$5.04	▼
 BTC 15 min · \$77,432.06 target			
Yes · Target Price: \$77,432.06	Cost \$ \$0.15	Apr 23 · Paid out \$0.19	▼
 BTC 15 min · \$77,429.64 target			
Yes · Target Price: \$77,429.64	Cost \$ \$1.98	Apr 23 · Paid out \$2	▼
 BTC 15 min · \$77,405.09 target			
Yes · Target Price: \$77,405.09	Cost \$ \$2.02	Apr 23 · Paid out \$2.04	▼
 BTC 15 min · \$77,405.66 target			
Yes · Target Price: \$77,405.66	Cost \$ \$1.06	Apr 23 · Paid out \$1.14	▼
 BTC 15 min · \$77,658.63 target			

Figure 10. Settled positions from the live session, taken from the Kalshi portfolio screen. The mix of small wins and small losses is consistent with a high-win-rate, low-magnitude maker strategy; round-trip P&L at the contract level is dominated by the spread captured on each side.

Two operational issues showed up immediately and would need to be hardened before scaling. First, the Kalshi REST API rate limit (~20 requests/second per user) is easy to brush against when actively quoting two sides in a fast market, particularly during cancel-replace storms; we implemented client-side throttling but did not yet have automatic backoff. Second, we briefly accumulated an unintended open position when the local order book state and Kalshi's server state desynchronized after a fill arrived between a cancel request and its response; this caused a position swing that consumed several minutes of P&L before being manually unwound. Both issues are tractable, but they are the obvious next priorities.

8. Conclusion and Future Work

Across the semester we built a complete pipeline from raw Kalshi WebSocket feeds through C++ book reconstruction, Python signal research, and live REST-based market making. We tested four progressively-refined strategies. OBI deltas as a market-taking signal: real but too small to clear 7% taker fees on a \$0.01 grid. Microprice: structurally unworkable because the calibrated bias is sub-tick on this contract. A gradient-boosted ensemble with IC ~0.28: predictive on a 10-second horizon but adversely selected as a market-maker entry filter. A simple rule-based maker (OBI threshold + BTC momentum sign): 91% win rate, +\$40 over 114 held-out sessions, validated in a 6-hour live trading run.

The headline finding is that the most profitable formulation we found is also the simplest. The directional information in the order book is real but small, and converting it into market-taking P&L on a fee-heavy binary market is hard. Once we re-cast the problem as a market-making one — where the edge comes from spread capture, \$0 maker fees, and the cancel-on-disagreement rule — the same OBI signal becomes consistently profitable. The ensemble model is an interesting object in its own right but, as a passive entry filter, was outperformed by a two-line rule.

Several extensions are natural follow-ups. The most operationally important is hardening the live system: robust position reconciliation on every fill message, automatic REST backoff under rate-limit pressure, and structured error handling for transient Kalshi API failures. On the strategy side, the spread compression we observed live suggests the best protection against newer makers is breadth: applying the same framework to other Kalshi markets with comparable activity (other crypto-anchored binaries, certain political event contracts during news windows) would diversify the edge. Finally, there is a richer arbitrage to investigate — the $\text{YES} + \text{NO} = \1 identity is sometimes briefly violated across the two sides of the book during fast moves, and a triangular taker would be a complementary, fee-paying but very short-horizon counterpart to our maker. None of these change the basic conclusion: on this market, simple beats clever.